STRUCTURAL TECHNICAL SPECIFICATION

Al Nakheel Tower — Package 3, Structural Works

Riyadh, Kingdom of Saudi Arabia

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B	10 Feb 2025	Revised following Employer & Contractor comments. Section 5.5 updated to reflect foundation waterproofing requirements. Mix design table updated. Issued for Construction (IFC).	M.A.K.	T.A.R.

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SECTION 1 — GENERAL

1.1 Scope of Work

This Specification establishes the minimum technical requirements for the design, procurement, supply, fabrication, installation, testing, and commissioning of all structural works to be executed by Gulf Consolidated Contracting WLL ("the Contractor") under Contract Reference ANRD/PKG3/STRUCT/2025-001 for the Al Nakheel Tower — Package 3 Structural Works ("the Works") in Riyadh, Kingdom of Saudi Arabia.

The Scope of Work under this Specification includes, but is not limited to:

- Excavation, blinding, and preparation of formation levels.
- Design, supply, and installation of all reinforced concrete foundations, including mat foundations, pile caps, and ground beams, as shown on the Structural Drawings referenced in Section 1.3.
- Supply, fabrication, and installation of all reinforcement steel for foundations, columns, shear walls, transfer slabs, post-tensioned suspended slabs, and other structural elements.
- Formwork design, supply, erection, and striking for all concrete elements.
- Supply, placement, and curing of all structural concrete in accordance with the mix design requirements set out in Section 3.1 of this Specification.
- Waterproofing systems to foundations and below-grade elements as specified in Sub-Section 5.5.
- Penetrations, blockouts, cast-in items, anchor bolts, and holding-down bolts as shown on the Drawings or instructed by the Engineer.
- All temporary works, shoring, propping, and re-propping required for the safe execution of the structural sequence.
- Testing, inspection, and quality control procedures as set out in Section 5.

This Specification shall be read in conjunction with the Structural Drawings (Reference Series ANT-STR-DWG-001 through ANT-STR-DWG-284 Rev. B), the FIDIC Conditions of Contract (Red Book, 1999 Edition), and the Particular Conditions forming part of the Contract. In the event of any conflict between this Specification and the Drawings, the Specification shall prevail unless otherwise directed in writing by the Engineer. Conflicts shall be immediately notified to the Engineer pursuant to Sub-Clause 4.1 of the Contract Conditions.

1.2 Applicable Standards and References

All Works shall be designed, procured, fabricated, installed, tested, and inspected in accordance with the latest editions (as at the Base Date of 1 December 2024) of the standards listed below, except as otherwise specified. Where this Specification imposes requirements more stringent than those of an applicable standard, the more stringent requirements shall govern. Nothing in the list of standards below shall relieve the Contractor of its obligations to comply with any applicable Saudi Government legislation, Royal Decree, or Ministry Circular issued before or after the Base Date.

1.2.1 Saudi Standards (SASO)

- SASO 143:2018 — Specification for Ordinary Portland Cement (equivalent to BS EN 197-1 CEM I 42.5N/R)
- SASO 1020:2018 — Ready-Mixed Concrete — Specification, Performance, Production and Conformity
- SASO 2789:2019 — Hot-Rolled Steel Bars and Rods for the Reinforcement of Concrete
- SASO GSO 11/2015 — Code of Practice for the Structural Use of Concrete
- Saudi Building Code SBC 301:2018 — Loads and Forces Requirements
- Saudi Building Code SBC 304:2018 — Concrete Structures Requirements

1.2.2 British / European Standards (BS EN)

- BS EN 206:2013+A2:2021 — Concrete: Specification, Performance, Production and Conformity
- BS EN 197-1:2011 — Cement: Composition, Specifications and Conformity Criteria for Common Cements
- BS EN 1992-1-1:2004+A1:2014 (Eurocode 2) — Design of Concrete Structures: General Rules
- BS EN 10080:2005 — Steel for the Reinforcement of Concrete — Weldable Reinforcing Steel

- BS EN 12350 (Parts 1–12) — Testing Fresh Concrete
- BS EN 12390 (Parts 1–12) — Testing Hardened Concrete
- BS EN 12504-1:2019 — Testing Concrete in Structures: Cored Specimens

1.2.3 ASTM International Standards

- ASTM A615/A615M-22 — Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
- ASTM A706/A706M-22 — Standard Specification for Low-Alloy Steel Deformed Bars for Concrete Reinforcement (where specified for seismic applications)
- ASTM C31 — Standard Practice for Making and Curing Concrete Test Specimens
- ASTM C39 — Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens
- ASTM C143 — Standard Test Method for Slump of Hydraulic-Cement Concrete
- ASTM C150 — Standard Specification for Portland Cement

1.2.4 American Concrete Institute (ACI)

- ACI 318-19 — Building Code Requirements for Structural Concrete and Commentary
- ACI 301-16 — Specifications for Structural Concrete
- ACI 305.1-14 — Specification for Hot Weather Concreting
- ACI 308R-16 — Guide to External Curing of Concrete

Where a standard has been superseded after the Base Date, the version applicable at the Base Date shall apply unless the Engineer instructs otherwise in writing.

SECTION 3 — CONCRETE WORKS

3.1 Concrete Mix Design

All structural concrete shall be produced by an approved ready-mix plant holding a current SASO product conformity certificate for concrete production. The Contractor shall submit concrete mix designs for the Engineer's review and approval not less than 28 days prior to the commencement of any concrete element. No concrete shall be poured without the Engineer's written approval of the mix design. Approved mix designs shall not be modified without the prior written consent of the Engineer.

The following table sets out the minimum concrete specification for each structural element. All concrete shall be designated concrete in accordance with BS EN 206 and SASO 1020. The characteristic compressive cylinder strength (fck) and characteristic cube strength (fck,cube) are given in the format C(fck)/(fck,cube).

Element / Location	Strength Class (BS EN 206)	Min. Cement Content (kg/m³)	Max. W/C Ratio	Min. Slump (mm)	Exposure Class
Foundation Slabs & Pile Caps	**C35/45**	350	0.45	100–160	XC2/XS1
Columns & Shear Walls (B1–Roof)	**C40/50**	380	0.40	120–160	XC1/XD1
Transfer Slabs & Post-Tensioned Floors	**C40/50**	380	0.40	160–200*	XC1
Suspended Slabs & Beams	**C35/45**	360	0.45	120–160	XC1
Slabs on Grade (Ground Floor)	**C30/37**	320	0.50	80–120	XC2
Blinding Concrete	**C15/20**	250	0.60	75–100	X0
Pre-cast Elements (where applicable)	**C40/50**	400	0.38	As mfr.	XC1/XD1

* Post-tensioned floors: target slump-flow 500–600 mm (self-compacting concrete may be specified by the Engineer). *NOTE: The concrete grades specified in this table represent absolute minima. Higher grades shall be used where required by the Structural Drawings, Structural Engineer's calculations, or the Engineer's written instruction.*

3.1.1 Cement and Supplementary Cementitious Materials (SCM)

All cement shall be Ordinary Portland Cement (OPC) conforming to SASO 143 or BS EN 197-1 CEM I 42.5N/R. Ground Granulated Blast-Furnace Slag (GGBS) or Pulverised Fuel Ash (PFA/Fly Ash) may be used as partial cement replacement subject to approval by the Engineer, provided that:

- GGBS replacement shall not exceed 50% by mass of total cementitious content.
- PFA replacement shall not exceed 25% by mass of total cementitious content.
- The total cementitious content (including SCM) shall satisfy the minimum values stated in the mix design table above.
- The use of SCM shall be demonstrated to meet the required characteristic strength by trial mixes and test results submitted for Engineer's approval.

3.1.2 Aggregates

Coarse aggregates shall be crushed limestone or crushed dolomite, clean and free from chlorides, sulphates, and deleterious substances, conforming to BS EN 12620. Maximum aggregate size shall be 20 mm for columns, shear walls, and other heavily reinforced elements, and may be 25 mm for foundation slabs and slabs on grade. Fine aggregates shall be washed natural sand or crushed fine aggregate conforming to BS EN 12620, with a fineness modulus between 2.3 and 3.1.

3.1.3 Hot Weather Concreting

Given the climatic conditions in Riyadh, the Contractor shall implement a Hot Weather Concreting Plan in accordance with ACI 305.1-14 and the Engineer's approval. The concrete temperature at point of delivery to the formwork shall not exceed 32°C. Where ambient temperatures exceed 35°C, the Contractor shall use chilled water, ice substitution, or liquid nitrogen cooling of aggregates, as approved by the Engineer. No concrete shall be poured when the ambient temperature exceeds 40°C without the Engineer's written permission and implementation of supplementary cooling measures.

3.4 Reinforcement

All reinforcement steel shall comply with the requirements of this Sub-Section 3.4, ASTM A615/A615M-22, SASO 2789, and BS EN 10080 as applicable.

3.4.1 Reinforcement Type and Grade

Unless otherwise stated on the Structural Drawings, all deformed reinforcement bars shall be:

- ASTM A615 Grade 60 deformed steel bars (fy = 420 MPa minimum; fu = 620 MPa minimum)
- Hot-rolled, conforming to SASO 2789 and ASTM A615/A615M-22
- Ribbed / deformed profile ensuring adequate bond with concrete
- Diameter range: T10 through T40 (metric sizes), as shown on Drawings

ASTM A706 Grade 60 low-alloy bars shall be used in all special moment-resisting frame elements and coupling beams where shown on the Structural Drawings, to ensure adequate ductility under seismic loading in accordance with SBC 304 and ACI 318-19 Chapter 18.

3.4.2 Concrete Cover to Reinforcement

Minimum concrete cover to all reinforcement (measured to the outer face of the outermost steel element, including links and ties) shall be as follows:

- Foundation slabs, pile caps, and ground beams: 50 mm minimum (measured to blinding)
- Columns (internal — above ground): 35 mm minimum
- Columns (external or exposed): 40 mm minimum
- Shear walls (internal): 35 mm minimum
- Shear walls (external or exposed to weather): 40 mm minimum
- Suspended slabs (top and soffit): 25 mm minimum
- Beams (internal): 30 mm minimum
- Slabs on grade: 50 mm minimum (from top face)

Cover shall be maintained using purpose-made plastic or concrete spacers/chairs of the correct size, fixed at maximum 800 mm centres for bars and 600 mm centres for mesh, and shall be of a type approved in advance by the Engineer. Wire chairs or twist-tie spacers shall not be used as cover spacers.

3.4.3 Splices and Laps

All lap lengths and splice lengths shall be as detailed on the Structural Drawings and in accordance with ACI 318-19 Table 25.5.2.1 and SBC 304. Mechanical couplers may be used as an alternative to lapping with the Engineer's prior written approval. Coupler systems shall be Type 2 conforming to ACI 318-19 Section 26.6.2 and certified to achieve 125% of the specified yield strength without failure.

SECTION 5 — QUALITY CONTROL متطلبات الجودة

The Contractor shall establish and maintain a Quality Management System (QMS) conforming to ISO 9001:2015 (or equivalent) for the full duration of the Works. The Contractor shall submit its Quality Plan, Inspection and Test Plans (ITPs), and Method Statements to the Engineer for approval not less than 21 days prior to commencement of each work activity. No structural element shall be commenced without approved ITPs in place and without the Engineer's written confirmation of readiness to proceed.

5.2 Testing Requirements — متطلبات الاختبار

All testing shall be performed by a SASO-accredited independent laboratory approved in advance by the Engineer. Testing shall not be carried out by a laboratory that has any commercial affiliation with the Contractor or any of its subcontractors. The Contractor shall bear all costs of testing, regardless of whether tests are passed or failed. The Engineer reserves the right to instruct additional tests at any time at the Contractor's cost if there are reasonable grounds for concern about materials or workmanship.

The minimum testing frequencies set out in the following table shall apply. The Contractor shall not reduce these frequencies without the Engineer's prior written approval:

Test Type	Standard	Frequency	Sample Size / Set	Acceptance Criteria
Cube Compressive Strength	BS EN 12390-3 / SASO 143	1 set per 50 m³ poured (min. 1 set/pour/day)	3 cubes per set (tested at 7, 28, 56 days)	fck,cube ≥ specified characteristic strength
Slump / Flow Test	BS EN 12350-2	Each truck / each pour start	1 sample per load	Within ±25 mm of target slump
Concrete Temperature	SASO / BS EN 206	Each truck	1 reading per load	8°C ≤ T ≤ 32°C at point of delivery
Air Content (where applicable)	BS EN 12350-7	1 per 100 m³	1 sample per batch	As mix design ± 0.5%
Core Testing (if cubes fail)	BS EN 12504-1	On Engineer's instruction	Min. 3 cores per area	Mean strength ≥ 0.85 × fck
Reinforcement Tensile Strength	ASTM A615 / SASO	Per heat number / delivery	3 bars per diameter per lot	fy ≥ 420 MPa; fu ≥ 620 MPa
Rebar Bend Test	ASTM A615	Per heat number	2 bars per diameter per lot	No fracture after 180° bend
Mill Certificate Review	ASTM A615 / BS EN 10080	Each consignment	All certificates	Comply with specified grade

NOTE: Where test results fail to meet the acceptance criteria, the Contractor shall immediately notify the Engineer and shall not continue placing concrete of the same mix design until the cause of failure has been identified and remediated to the Engineer's satisfaction. A Non-Conformance Report (NCR) shall be raised within 24 hours of a failed test result and submitted to the Engineer for review.

5.2.1 Concrete Cube Testing — Detailed Requirements

Concrete cube testing shall be performed as follows, in addition to the minimum frequencies stated in the table above:

- **i.** Samples shall be taken from the concrete discharged at the point of final placement in the formwork, not from the mixer truck drum.
- **ii.** A minimum of one (1) set of three (3) cube samples shall be taken per 50 m³ of concrete poured, or at least one set per individual pour irrespective of volume, whichever is the more frequent.
- **iii.** Each set shall consist of three (3) cubes of 150 mm × 150 mm × 150 mm, moulded, cured, and tested in accordance with BS EN 12390-1 and BS EN 12390-3.
- **iv.** One cube from each set shall be tested at 7 days (indicative result), one at 28 days (acceptance result), and one retained and tested at 56 days (confirmation).
- **v.** Acceptance of concrete shall be based on the 28-day characteristic compressive cube strength meeting the criteria of BS EN 206 Clause 8.2.1.

vi. All cube samples, test records, and test certificates shall be submitted to the Engineer within 48 hours of each test result being available.

5.2.2 Reinforcement Testing

In addition to the review of mill certificates, the Contractor shall carry out independent physical testing of all reinforcement delivered to Site. Testing shall be carried out on samples taken from each consignment (each delivery constituting a separate heat or melt number) in the presence of the Engineer's Representative. Test results shall be submitted within 5 working days of sampling. No reinforcement shall be incorporated into the Works unless its mill certificate has been reviewed and physical testing results confirm conformance with ASTM A615 Grade 60 and SASO 2789.

5.5 Foundation Works — أعمال الأساسات

SPECIFICATION NOTE — CONFLICT WITH SECTION 3.1: Sub-Section 5.5.1 of this Section specifies a minimum concrete grade of C40/50 for foundation slabs. This conflicts with Section 3.1, which specifies C35/45 for foundation slabs and pile caps. In accordance with the document hierarchy set out in Sub-Clause 1.5 of the Contract, the more onerous requirement (C40/50 per Sub-Section 5.5.1) shall prevail until the discrepancy is resolved by the Engineer's written clarification. The Contractor shall notify the Engineer immediately pursuant to Sub-Clause 4.1 of the Contract.

5.5.1 Foundation Concrete Grade

Foundation slabs, mat foundations, pile caps, and all below-grade structural concrete elements shall use a minimum concrete grade of **C40/50** (characteristic cylinder strength fck = 40 MPa; characteristic cube strength fck,cube = 50 MPa) in accordance with BS EN 206, to achieve enhanced durability performance in the aggressive ground conditions identified in the Geotechnical Investigation Report (Reference ANRD/GEO/2024-004) for the Al Nakheel Tower Site.

This requirement reflects the sulphate and chloride exposure conditions identified in the geotechnical investigation and supersedes the general mix design table in Section 3.1 for foundation elements. The minimum cement content for this C40/50 foundation concrete shall be 380 kg/m³ and the maximum water-to-cement ratio shall be 0.40.

5.5.2 Foundation Waterproofing — Crystalline Admixture

All below-grade concrete elements, including mat foundations, pile caps, retaining walls, and basement walls to depth B1 level, shall incorporate a crystalline waterproofing admixture to provide integral, self-healing waterproofing protection. The crystalline admixture shall comply with the following requirements:

- Type: Crystalline Admixture Type 1 — integral concrete admixture (powder or liquid) to be added to the concrete mix at the batching plant, in accordance with the manufacturer's recommendations and the Engineer's approval.
- Standard: ASTM C1202-22 (Rapid Chloride Permeability Test) — the admixture-treated concrete shall demonstrate a permeability rating of 'Very Low' (charge passed < 1000 coulombs at 28 days) as tested in accordance with ASTM C1202.
- Dosage: As per manufacturer's current published data sheets, typically 1.0–2.0% by weight of cement, confirmed by trial mix.
- Approved products (subject to Engineer's written confirmation): Xypex Admix C-Series, Kryton KIM, Penetron Admix, or equivalent — equivalence to be demonstrated by submission of technical data, test certificates, and third-party performance data for review by the Engineer.

The crystalline admixture shall be used in addition to, and not as a replacement for, any external waterproof membrane system shown on the Architectural or Structural Drawings. Where both an external membrane and a crystalline admixture are specified, both shall be provided.

5.5.3 Pre-Pour Inspection and Sign-Off

The Contractor shall not commence placing concrete for any foundation element without having obtained the Engineer's written sign-off on the Pre-Pour Inspection Checklist (Form QC-03, attached as Appendix F). The checklist shall confirm, as a minimum:

a. Formation level and bearing capacity verified by Engineer's Representative.

b. Blinding concrete placed, cured, and inspected.
c. Waterproofing system (where applicable) inspected and approved.
d. Reinforcement — type, grade, size, spacing, laps, and cover — inspected and approved.
e. Cast-in items, anchor bolts, sleeves, and penetration blockouts positioned and secured.
f. Formwork inspected for line, level, stability, and cleanliness.
g. Approved concrete mix design and delivery docket review procedure confirmed.
h. Testing laboratory mobilised and sampling equipment on Site.

— End of Specification Excerpt —

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